Pain Management Non Cardiac/
Procedural Analgesia
Effective: July 10, 2015
Revised: June 10, 2024
G-5

Inclusion criteria: Patients complaining of severe pain. This protocol may also be utilized during extrication, or for procedures such as cardioversion and transcutaneous pacing.

DO NOT IMPLEMENT PAIN MANAGEMENT PROTOCOL TO TREAT HEADACHES/MIGRAINES.

Determine pregnancy status. DO NOT IMPLEMENT PAIN MANAGEMENT FOR LABOR. Provide BLS pain control measures first (positioning, splinting, ice, traction, etc.) before initiating this protocol.

Patients may display a wide variation of response to opioid medications. Minimal doses of opioids may cause respiratory depression in the elderly and possibly intoxicated patients.

Ketamine is an NMDA antagonist that has both pain control (at low doses < 0.5-1 mg/ kg) and dissociative effects. Ketamine has benefits when compared to typical opioid analgesics (morphine, fentanyl) in that ketamine does not cause respiratory depression generally, even at high doses compared to opioids, and may cause a modest increase in blood pressure and heart rate (generally thought of as favorable in the significantly injured patient). Ketamine also retains effectiveness even in the opioid-tolerant patient by working through a different receptor.

- Generally speaking, ketamine is a relatively safe medication, with fewer severe side effects than opioids in many cases (2,3). Side effects of ketamine include hypertension, increased secretions, and nausea/ vomiting. Pretreatment with anti-emetics when giving ketamine is advised.
- Patients may experience “emergence reactions” as ketamine wears off, indicated by significant agitation which, while temporary, can be difficult to control.
- Ketamine also can cause increased intraocular pressure and should be avoided in cases of suspected acute angle closure glaucoma
- It may cause increased heart rate and increased myocardial demand and should not be used in cases of suspected myocardial infarction (i.e. chest pain).
- Laryngospasm has been reported with ketamine use (usually at dissociative (>1 mg/kg) doses, and is generally manifested by stridor, apnea, or hypoxia.

Acetaminophen acts on the hypothalamus to produce antipyresis. It may work peripherally to block pain impulse generation; may also inhibit prostaglandin synthesis in the CNS.

Begin Initial Medical Care.

Pain severity (0-10) is a vital sign that must be recorded before and after IV or IN administration of pain medication. Vital signs should be obtained before and after all doses of pain medication delivery.

In the pediatric patient and in those that cannot communicate, utilize the Wong Baker Faces Pain Rating Scale.

Patients receiving pain medication/sedation should also receive SpO2 monitoring. Capnography and cardiac monitor may be placed. If the adult or pediatric patient is experiencing a sickle cell crisis, administer 10ml/kg normal saline, then KVO.

ADULT Pain management for non cardiac applications and extrication:

REFER TO CHEST PAIN PROTOCOL FOR SPECIFIC PAIN MANAGEMENT ASSCOCIATED WITH CARDIAC PATIENTS

Fentanyl 0.5-1.0 mcg/kg (Max single dose 100 mcg) IVP slowly or IN (25-50 mcg per nostril) for severe pain if:

- GCS ≥ 14
- Pain scale > 5
- B/P ≥ 90 systolic

Fentanyl may be repeated at a dose of 50 mcg every 10 minutes as long as the above parameters are still met. Additional doses prior to a 10 minute interval are **AT MD DISCRETION ONLY.**

OR

Administer **0.2mg/Kg ketamine** IV slow push over 90 seconds.

May repeat **at 0.2 mg/Kg ketamine** in 10 minutes if:

- GCS ≥ 14
- Pain scale > 5
- B/P ≥ 90 systolic

Contraindications to pain management with Fentanyl or Ketamine:

- GCS < 14
- Allergic to Fentanyl or Ketamine
- Isolated headache
- Labor pain
- Suspected acute angle closure glaucoma

Contraindication specific to Ketamine:

- Suspected acute angle closure glaucoma

Precautions when administering Ketamine:

- Laryngospasm
  i) Apply very firm pressure bilaterally to the notch between the patient's mandibular condyle and mastoid process.
  ii) If not immediately effective, initiate positive pressure ventilation with a BVM
  iii) If not resolved, administer 2mg Midazolam IVP every 3minutes and prepare for aggressive airway management.
- Emergence Reaction:
  i) Administer 2-5 mg Midazolam slow IVP
- Nausea Vomiting
  ii) Administer 4 mg Zofran slow IVP as needed

Acetaminophen may be administered solely or in combination with Fentanyl **or** Ketamine

Administer 1 gram Tylenol IV/IO over 15 minutes.

Contraindications to Acetaminophen:

- Allergy to acetaminophen
- Severe active hepatic disease

PEDIATRIC Pain management/Extrication: Dosing for child that cannot communicate is AT MD DISCRETION ONLY.

Adminisdter Fenatnyl 1 mcg/Kg(Maximum single dose 50mcg) slow IVP or IN (half dose in each nostril) if:

Child can communicate and understand Wong Baker Scale.
Pain scale > 5
B/P ≥ to age appropriate

May be repeated every 10 minutes as long as the above parameters are still met.
Additional doses required before the 10 minute interval are AT MD DISCRETION ONLY.

OR

Administer **0.2mg/Kg ketamine** IV slow push over 90 seconds.
May repeat **at 0.2 mg/Kg ketamine** in 10 minutes if:
Infant or child continues to communicate and understand the Wong Baker Faces with pain > 5 on 1-10 pain scale that has a B/P > the age appropriate.

Contraindications to pain management with Fentanyl or Ketamine:

- Infant or child cannot communicate or understand the Wong Baker faces
- Allergic to Fentanyl or Ketamine
- Complaint of non-traumatic chest pain
- Isolated headache

Precautions when administering Ketamine:

- Laryngospasm
  i) Apply pressure as noted in the adult
  ii) If not effective, initiate positive pressure ventilation with a BVM
  iii) If not resolved, administer 0.2 mg/kg midazolam IVP every 5 minutes and prepare for aggressive airway management. Maximum single dose is 5 mg. Maximum cumulative dose is 1 mg/kg or 20 mg.
- Emergence Reaction: As ketamine wears off, indicated by significant agitation which, while temporary, can be difficulty to control.
  i Administer 0.2 mg/kg midazolam IVP over 2 minutes. Maximum single dose is 5 mg.

Acetaminophen may be administered solely or in combination with Fentanyl **or** Ketamine

Administer Acetaminophen 15mg/kg over 15 Minutes

Contraindications to Acetaminophen:

- Allergy to acetaminophen
- Severe active hepatic disease

Adult Pain Management for Synchronized Cardioversion:
DO NOT DELAY PRUDENT CARE FOR SEDATION IN THE SEVERELY COMPROMISED PATIENT.
If sedation needed:
Versed: **10** mg midazolam IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP if: GCS ≥ 14

Pediatric Pain Management for Synchronized Cardioversion:
AT MD DISCRETION ONLY

Adult Pain Management for Transcutaneous Pacing

INITIATE PACING FIRST. NEVER DELAY NEEDED PACING FOR PRE-EMPTIVE PAIN CONTROL/SEDATION.

Fentanyl 0.5-1.0 mcg/kg (Max single dose 100 mcg) slow IVP or IN (25mcg per nostril) for pain control if:

GCS ≥ 14
Pain scale > 5
B/P ≥ 90 systolic

Fentanyl may be repeated at 50 mcg every 10 minutes as long as the above parameters are still met.

Additional doses required before the 10 minute interval are AT MD DISCRETION ONLY.

OR

Versed: **10** mg midazolam IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP if:

GCS ≥ 14

Pediatric Pain Management for Transcutaneous Pacing

AT MD DISCRETION ONLY